Network Monitoring Lab — Zabbix 7.0

Home Lab Documentation | Ubuntu Server 24.04 | Zabbix 7.0 LTS | Daksh Patel

1. Project Overview

This lab deploys Zabbix 7.0 LTS on Ubuntu Server 24.04 to implement enterprise-grade network and infrastructure monitoring. Zabbix monitors the DNS-DHCP server built in Project 2, collecting 77+ live metrics including CPU utilization, memory usage, disk space, and network interface statistics. A custom dashboard visualizes all metrics in real time with alerting configured for critical thresholds.

Lab Environment

Monitoring Tool	Zabbix 7.0 LTS
Server OS	Ubuntu Server 24.04.4 LTS (aarch64)
Database	MySQL (mysql-server)
Web Server	Apache2
Zabbix Server IP	192.168.56.2
Web Interface	http://192.168.56.2/zabbix
Monitored Hosts	2 — DNS-DHCP-SERVER + Zabbix server itself
Total Metrics	77+ items on DNS-DHCP-SERVER, 128 on Zabbix server
Agent Type	Zabbix Agent (passive) — port 10050
Template Used	Linux by Zabbix agent

2. Metrics Collected

Category	Metric	Live Value	Status
CPU	CPU utilization	6.21%	Normal
CPU	CPU idle time	93.78%	Normal
CPU	Context switches/sec	1091.87	Normal
Memory	Available memory	811.6 MB	Normal
Memory	Memory utilization	58.3%	Normal
Storage	FS [/]: Space available	18.55 GB	Normal

Storage	FS [/]: Space total	23.4 GB	Normal
Storage	FS [/]: Inodes free	93.11%	Normal
Security	Checksum of /etc/passwd	Monitored	Normal
Alert	High CPU trigger	>90% threshold	Configured

3. Installation Commands

Install Zabbix Repository

```
wget https://repo.zabbix.com/zabbix/7.0/ubuntu/pool/main/z/zabbix-release/zabbix-release_latest+ubuntu24.04_all.deb
sudo dpkg -i zabbix-release_latest+ubuntu24.04_all.deb
sudo apt update
```

Install Zabbix Components

```
sudo apt install zabbix-server-mysql zabbix-frontend-php zabbix-apache-conf
zabbix-sql-scripts zabbix-agent -y
```

Configure Database

```
sudo mysql
CREATE DATABASE zabbix CHARACTER SET utf8mb4 COLLATE utf8mb4_bin;
CREATE USER 'zabbix'@'localhost' IDENTIFIED BY 'zabbix';
GRANT ALL PRIVILEGES ON zabbix.* TO 'zabbix'@'localhost';
FLUSH PRIVILEGES;
zcat /usr/share/zabbix-sql-scripts/mysql/server.sql.gz | mysql
--default-character-set=utf8mb4 -uzabbix -p zabbix
```

Start Services

```
sudo systemctl restart zabbix-server zabbix-agent apache2
sudo systemctl enable zabbix-server zabbix-agent apache2
```

4. Verification — Live Dashboard

Custom Zabbix dashboard showing all 5 widgets — Problems panel (no active alerts), CPU utilization graph (5–7%), Memory utilization graph (57.6–58.6%), Host availability table (2 hosts, all green), and live clock.

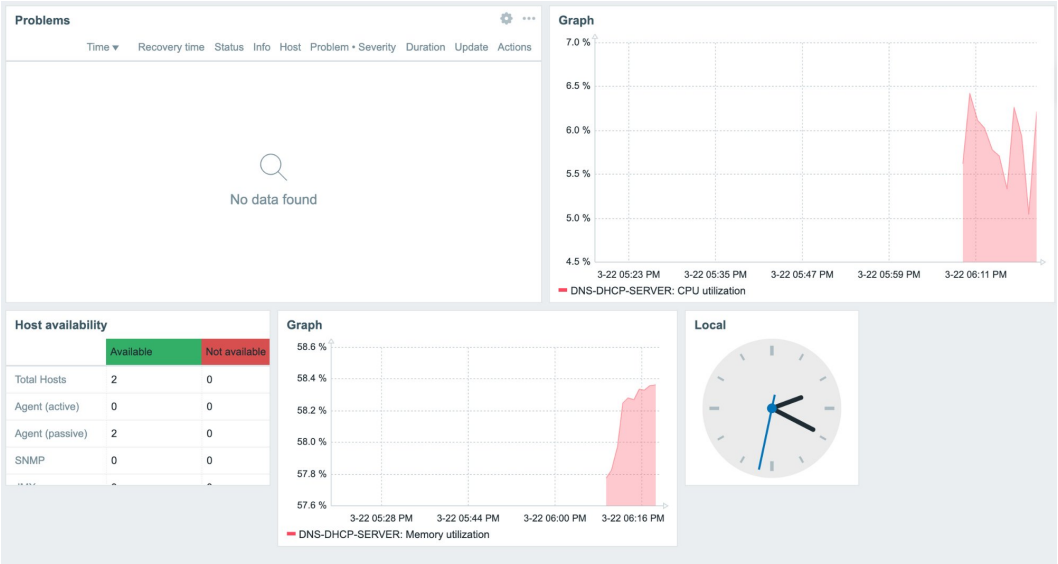

Figure 1: Lab Monitoring Dashboard — CPU, Memory, Host Availability, Problems, Clock

5. Verification — Latest Data (77 Metrics)

Latest data view showing 77 live metrics collected from DNS-DHCP-SERVER — including CPU idle at 93.78%, memory at 811.6MB available, disk space 18.55GB free, context switches, iowait, and filesystem statistics all updating in real time.

Host	Name ▲	Last check	Last value	Change	Tags	
DNS-DHCP-SERVER	Available memory	20s	811.6 MB	-672 KB	component: memory	Graph
DNS-DHCP-SERVER	Available memory in %	19s	41.5793 %	-0.05178 %	component: memory	Graph
DNS-DHCP-SERVER	Checksum of /etc/passwd	10m 22s	d34f4aa5853852…		component: security	History
DNS-DHCP-SERVER	Context switches per second	44s	1091.8714	+65.3576	component: cpu	Graph
DNS-DHCP-SERVER	CPU guest nice time	42s	0 %		component: cpu	Graph
DNS-DHCP-SERVER	CPU guest time	43s	0 %		component: cpu	Graph
DNS-DHCP-SERVER	CPU idle time	41s	93.7808 %	-1.1656 %	component: cpu	Graph
DNS-DHCP-SERVER	CPU interrupt time	40s	0 %		component: cpu	Graph
DNS-DHCP-SERVER	CPU iowait time	39s	0.8328 %	+0.01602 %	component: cpu	Graph
DNS-DHCP-SERVER	CPU nice time	38s	0 %		component: cpu	Graph
DNS-DHCP-SERVER	CPU softirq time	37s	0.05099 %	-0.000035 %	component: cpu	Graph
DNS-DHCP-SERVER	CPU steal time	36s	0 %		component: cpu	Graph
DNS-DHCP-SERVER	CPU system time	35s	2.4813 %	+0.2364 %	component: cpu	Graph
DNS-DHCP-SERVER	CPU user time	34s	2.9747 %	+0.8655 %	component: cpu	Graph
DNS-DHCP-SERVER	CPU utilization	41s	6.2192 %	+1.1656 %	component: cpu	Graph
DNS-DHCP-SERVER	Free swap space	28s	0 B		component: memory component: storage	Graph
DNS-DHCP-SERVER	Free swap space in %	27s	100 %		component: memory component: storage	Graph
DNS-DHCP-SERVER	FS [/]: Get data	21s	{"fsname":"/","op…		component: raw component: storage filesystem: / ···	History
DNS-DHCP-SERVER	FS [/]: Inodes: Free, in %	21s	93.1185 %		component: storage filesystem: / fstype: ext4	Graph
DNS-DHCP-SERVER	FS [/]: Option: Read-only	21s	0		component: storage filesystem: / fstype: ext4	Graph
DNS-DHCP-SERVER	FS [/]: Space: Available	21s	18.55 GB	-64 KB	component: storage filesystem: / fstype: ext4	Graph
DNS-DHCP-SERVER	FS [/]: Space: Total	21s	23.4 GB		component: storage filesystem: / fstype: ext4	Graph
DNS-DHCP-SERVER	FS [/]: Space: Used	21s	3.64 GB	+64 KB	component: storage filesystem: / fstype: ext4	Graph

Figure 2: Latest data — 77 live metrics from DNS-DHCP-SERVER updating every 20–40 seconds

6. Verification — Host Status

Hosts page confirming both monitored hosts show green ZBX availability status — DNS-DHCP-SERVER (77 items, 15 graphs) and Zabbix server (128 items, 10 graphs) both Enabled with agent responding on 127.0.0.1:10050.

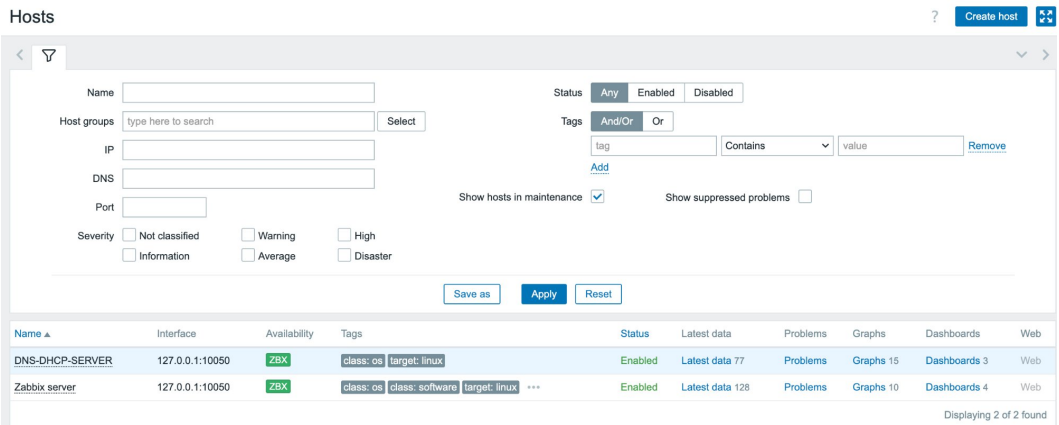

Figure 3: Hosts page — DNS-DHCP-SERVER and Zabbix server both ZBX green, Enabled

7. Verification — CPU Utilization Graph

Full CPU utilization graph for DNS-DHCP-SERVER showing historical data — last 5.40%, min 5.05%, avg 5.84%, max 6.43%. Trigger configured to alert when CPU exceeds 90% threshold.

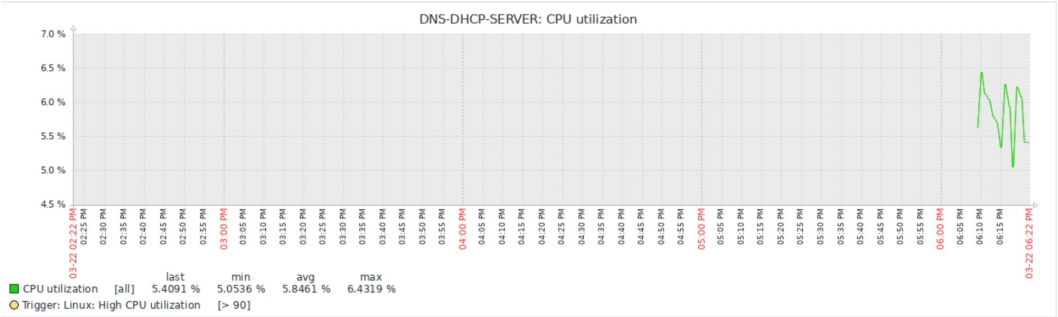

Figure 4: DNS-DHCP-SERVER CPU utilization — last 5.40%, avg 5.84%, trigger >90%

8. Skills Demonstrated

Zabbix	Server installation, web setup, host configuration, dashboard creation
Monitoring	77+ metrics collected — CPU, memory, disk, network, security
Alerting	Trigger configured for High CPU (>90%) threshold
MySQL	Database creation, user provisioning, schema import
Apache	Web server configuration for Zabbix frontend
Linux Admin	Service management, package installation, systemctl
SNMP	Agent-based monitoring via Zabbix agent port 10050
Dashboard	Custom 5-widget dashboard with real-time graphs and alerts
Troubleshooting	MySQL socket error, user permissions, template import issues resolved